FLAT PULLBACK ALONG TRANSLATION IS A FLAT MORPHISM OF GROUP SCHEMES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Lemma morphisms-lemma-base-change-flat (canonical tag 01U9). cited at proof lines 849-850. Flatness is stable under base change.

Source reference. Original lemma 01U9: sources/stacks-chow/tags.txt, source lines 2402–2402. Cited at proof lines 849-850: the flat map T -> S pulls back along the cartesian square to the action morphism; load-bearing.

Lemma 0.2. *Let S be a scheme. Let G be a group scheme over S. Let T be a scheme over S and let $\psi : T \to G$ be a morphism over S. If T is flat over S, then the morphism*

$$T \times_S G \longrightarrow G, \quad (t, g) \longmapsto m(\psi(t), g)$$

is flat. In particular, if G is flat over S, then $m : G \times_S G \to G$ is flat.

Proof. Consider the diagram

$$\begin{array}{ccccc} T \times_S G & \xrightarrow[(t,g)\mapsto(t,m(\psi(t),g))]{} & T \times_S G & \xrightarrow{\mathrm{pr}} & G \\ & & \downarrow & & \downarrow \\ & & T & \longrightarrow & S \end{array}$$

The left top horizontal arrow is an isomorphism and the square is cartesian. Hence the lemma follows from Morphisms, Lemma 01U9. □

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